SOFTWARE ENGINEERING
LAB-1

1.FUNCTIONAL REQUIREMENTS

				Acceptance Criteria	**Rationale**
FR-001	Functional (Supply Allocation)	The system shall match incoming field shelter supply requisitions against central warehouse inventory and assign delivery priority based on criticality (e.g., medical kits, water).	High	**Pass:** Critical supplies are prioritized and assigned to the dispatch manifest first. **Fail:** Low-priority items are dispatched before critical needs.	Prevents life-threatening shortages by automating triage-based resource distribution.
FR-002	Functional (Volunteer Dispatch)	The system shall register volunteer skill sets and	High	**Pass:** Certified medical volunteers are assigned to triage zones	Ensures effective on-ground response by aligning critical disaster

				Acceptance Criteria	Rationale
		match volunteers to specific shelter tasks or rescue zones based on required expertise.		rather than general logistics. **Fail:** Unqualified personnel are assigned to specialized tasks.	tasks with qualified personnel.
FR-003	Functional (Fleet Tracking)	The system shall track relief vehicle locations and delivery status in real-time via GPS coordinates.	Medium	**Pass:** Vehicle position updates on the manager dashboard within 30 seconds of movement. **Fail:** In-transit vehicles show stale location data exceeding 5 minutes.	Provides visibility into supply chain bottlenecks and ensures accountability of relief goods.
FR-004	Function	The	High	**Pass:**	Enables

				Acceptance Criteria	Rationale
	al (Shelter Mapping)	system shall provide an interactive GIS map displaying shelter capacities, current supply levels, and reported shortage alerts.		Shelter status indicators change color dynamically when stock drops below threshold. **Fail:** Dashboard fails to reflect urgent shelter depletion flags.	disaster managers to make data-driven decisions during rapid field changes.
FR-005	Functional (Offline Sync)	The mobile client shall allow relief volunteers to log fulfilled distributions offline and synchronize data once	Medium	**Pass:** Cached offline logs successfully upload to the central database upon reconnecting. **Fail:** Data logged offline is overwritten	Disaster-struck areas frequently experience cellular outages; field workers must still capture records.

				Acceptance Criteria	Rationale
		network connectivity resumes.		or lost after reconnection.	

2.NON FUNCTIONAL REQUIREMENTS

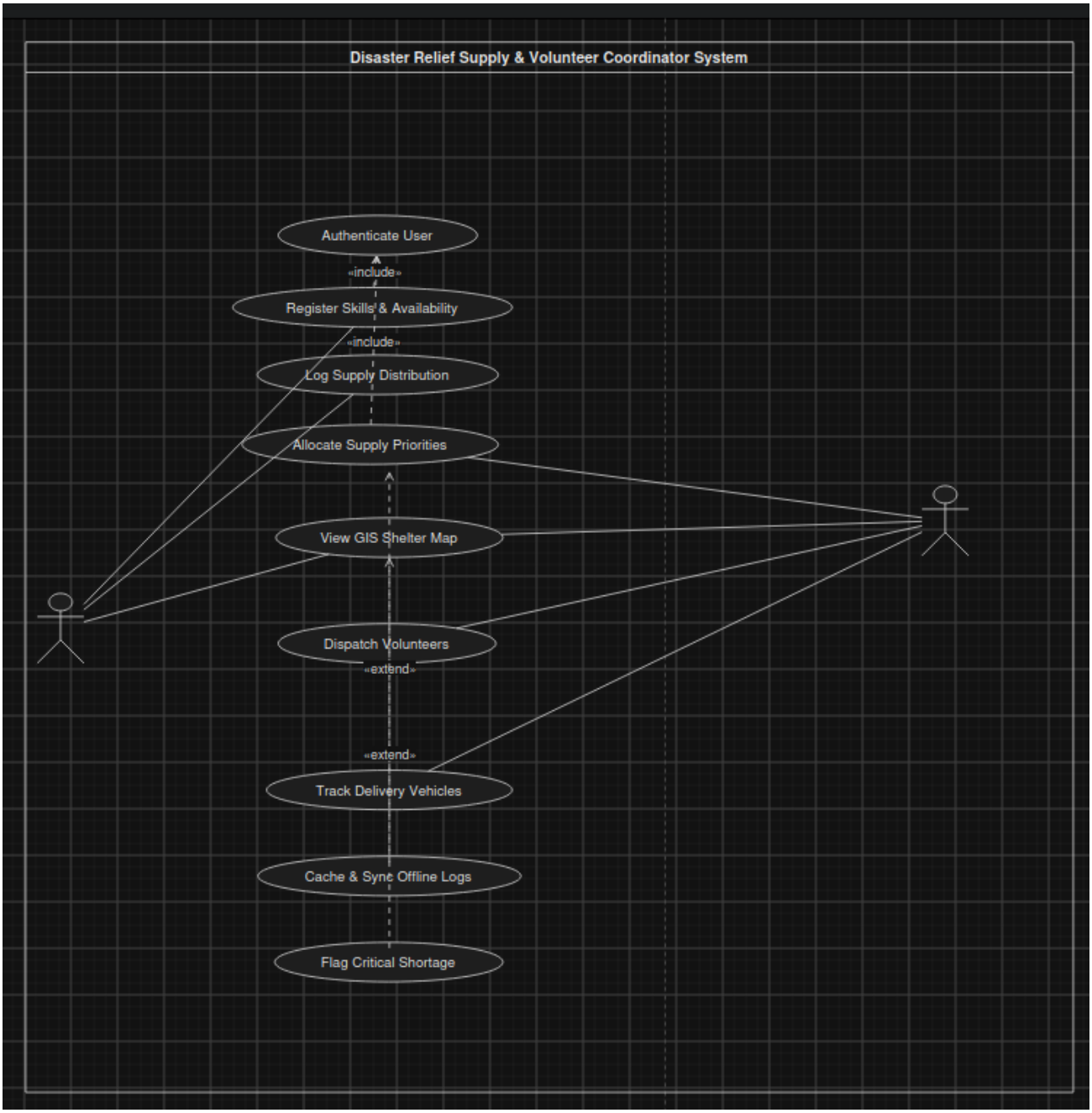
Requirement ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Offline Resilience	The disaster mapping dashboard must operate under low network bandwidth conditions and support offline GIS tile caching.	High	**Pass:** Map renders cached layers within 2 seconds on 2G/low-bandwidth connections. **Fail:** Dashboard stalls or fails to load without high-speed internet.	Critical disaster response operations must continue without disruption in areas with damaged infrastructure.
NFR-002	Security & Data Protection	The system must enforce	High	**Pass:** Unauthorized users are	Protects sensitive identity

Requirement ID	Type	Description	Priority	Acceptance Criteria	Rationale
		Role-Based Access Control (RBAC) and encrypt all shelter and volunteer data at rest and in transit using TLS 1.3 and AES-256.		blocked from administrative controls and all intercepted traffic is encrypted. **Fail:** Sensitive volunteer PII or supply logistics data is accessible in plaintext.	data and prevents malicious disruption of relief supply routes.
NFR-003	Availability & Reliability	The central management platform must maintain an uptime of 99.9% during active disaster response operations, with automatic failover to redundant	High	**Pass:** Unscheduled downtime does not exceed 43 minutes per month, and failover completes in under 30 seconds. **Fail:** Platform crashes during an	System unavailability during an ongoing crisis directly delays critical life-saving relief dispatches.

Requirement ID	Type	Description	Priority	Acceptance Criteria	Rationale
		cloud servers.		active emergency and requires manual restart.	
NFR-004	Scalability & Concurrency	The system shall handle up to 10,000 concurrent active users (volunteers and coordinators) and process up to 1,000 requisitions per minute without service degradation.	Medium	**Pass:** Response time remains under 1.5 seconds under peak simulated load. **Fail:** Requests time out or throw 5xx errors during sudden mass-casualty event surges.	Rapid influxes of volunteers and shelter requests during large-scale disasters must not crash the backend.
NFR-005	Usability & Accessibility	The user interface for field volunteers must	Medium	**Pass:** 90% of novice volunteers complete an intake/distri	Volunteers frequently work in harsh field conditions

Requirement ID	Type	Description	Priority	Acceptance Criteria	Rationale
		adhere to WCAG 2.1 Level AA standards, featuring high-contrast modes and support for single-handed mobile operation.		bution report in under 60 seconds during field trials. **Fail:** Form fields are unreadable in direct sunlight or require complex multi-step navigation.	(e.g., bright sunlight, extreme weather) and need rapid, frictionless data entry.

UML DIAGRAM



Use-Case Flow Specification

Use Case Name: Match and Allocate Shelter Supplies

Use Case ID: UC-001

Primary Actor: Disaster Manager

Supporting Actors: Central Warehouse Inventory System, Field Shelter Coordinator

1.Brief Description The Disaster Manager reviews incoming shelter supply requests (e.g., water, medical kits, blankets), matches them against available warehouse stock, prioritizes dispatch orders based on triage urgency, and generates a delivery manifest for logistics routing.

2. Preconditions

- he Disaster Manager is authenticated into the system.
- t least one shelter requisition exists in the system queue.
- warehouse inventory levels are loaded and accessible.

3. Postconditions

- upply allocations are committed to the central database.
- arehouse inventory levels are decremented accordingly.
- prioritized dispatch manifest is generated and assigned to a delivery vehicle.

4. Main Success Scenario (MSS)

1. The Disaster Manager opens the **Supply Allocation** module.
2. The system retrieves and displays all pending shelter requisitions sorted by urgency rating (Critical, High, Medium, Low).
3. The Disaster Manager selects a critical shelter requisition for processing.
4. The system queries warehouse inventory and matches requested items against real-time stock levels.
5. The system automatically assigns highest dispatch priority to life-critical items (water, medical kits).
6. he Disaster Manager confirms the allocation and selects an available relief delivery vehicle.
7. he system updates the warehouse stock count, generates the dispatch manifest, and sets the requisition status to Dispatched.
8. he system displays a confirmation banner and routes the manifest to the vehicle tracking dashboard.

5. Alternate Flow

- **AF-1: Insufficient Warehouse Stock (Partial Allocation)**
    - **rigger:** Step 4 indicates available warehouse stock is less than the requested quantity.
    - **Flow:**
        1. The system flags the item as Understocked and calculates the maximum allocatable quantity.

2. The system prompts the Disaster Manager to choose: *(a) Partial Dispatch* or *(b) Re-route from Secondary Warehouse*.
3. The Disaster Manager selects *Partial Dispatch*.
4. The system allocates all available units to the current manifest and creates a split backorder requisition for the remaining deficit.
5. The system resumes from **Step 6** of the Main Success Scenario.

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github: https://github.com/vasanthreddy121/SE-LAB1.git